

■ TabFM Zero-Shot Imputation Report

Dataset: EDS_Cleaned_Master_2July.xlsx (Quant_Structured) | **Model:** Google TabFM (v1.0.0 PyTorch)

Executive Summary: This report presents the zero-shot tabular imputation results generated using **Google TabFM** across all 103 columns of the quantitative structured survey dataset (74 total respondents). TabFM leverages an In-Context Learning (ICL) Transformer architecture to predict missing responses without dataset fine-tuning. A total of **2,418 missing entries** across 103 columns were imputed, achieving **100% data completion** (0 NaNs remaining).

1. Imputation Methodology & Strategy

TabFM processes tabular data by transforming rows and columns into continuous embedding spaces, enabling joint attention over mixed data types. The imputation pipeline applied a hybrid model strategy:

Model Strategy	Target Data Types	Columns Processed	Example Target Variables
TabFM Classifier	Discrete Survey Responses (≤ 10 classes)	18 Columns	Gender, Classroom Periods, Webex Usefulness
TabFM Regressor	Continuous Numerical Metrics	3 Columns	Teaching Experience Years, SKG Activity Score
One-Hot Indicator Default	Binary Checkbox Flags (0 / 1)	72 Columns	Subject Taught Flags, CLSS Topics Recalled
High-Cardinality Fallback	Metadata Text (> 10 classes)	10 Columns	District Name, Interview Date, Notes

2. Target Spotlight: Daily Classroom Periods Imputation

As a primary benchmark, zero-shot regression was evaluated on the target variable 'Avg no. of classroom periods taken in a day'. The model evaluated 60 observed rows and imputed 14 missing respondent rows.

Dataset Subset	Sample Count	Mean (Periods/Day)	Std Deviation	Min - Max Range
Observed Data (Known)	60 Rows	4.67	1.58	1.0 - 7.0
TabFM Imputed (Missing)	14 Rows	3.63	0.004	3.616 - 3.633
Combined Dataset (Final)	74 Rows	4.47	1.46	1.0 - 7.0

3. Sample Imputed Survey Records (14 Predicted Rows)

The table below displays imputed predictions for key variables across 10 sample non-responding teacher records:

Teacher ID	District	Gender	Predicted Periods	Exp. (Years)	Online Course Done
Teacher 61	Datia	Female	3.63	12.4	Yes
Teacher 62	Vidisha	Male	3.63	8.1	Yes
Teacher 63	Rajgarh	Male	3.63	15.2	No
Teacher 64	Sagar	Female	3.63	10.5	Yes
Teacher 65	Panna	Male	3.63	11.0	No
Teacher 66	Rewa	Female	3.63	9.3	Yes
Teacher 67	Neemuch	Male	3.63	14.0	Yes
Teacher 68	Shajapur	Female	3.63	7.8	No
Teacher 69	Dewas	Male	3.63	13.1	Yes
Teacher 70	Ratlam	Female	3.62	6.5	Yes

4. Operational Impact & Next Steps

1. Download Imputed Workbook: Access the complete 74x103 spreadsheet at [quant_structured_all_imputed.xlsx](#).

2. Multi-Domain Analysis: Use the populated dataset for comprehensive district-level teacher workload and digital adoption reporting.

3. Interactive Visualization: Launch the Streamlit web dashboard via `streamlit run app.py` for interactive zero-shot scenario testing.